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# Sum-throughput maximization for UAV-aided energy-harvesting WBANs: Hovering altitude design, power control and time allocation☆

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ABSTRACT

UAV-aided energy-harvesting WBANs where sensor nodes can harvest RF energy transmitted by a UAV have attracted considerable attention. However, how to maximize the sum-throughput of the sensor nodes with cross-tier interferences from satellite and cellular systems has not been addressed. In order to bridge this gap, we devote this paper to developing two schemes with the fixed or variable duration of the energy-harvesting phase. Specifically, we formulate two non-convex optimization problems corresponding to the two schemes, and transform them to tractable subproblems, and accordingly two algorithms are proposed named HA-PC-TA and UAVPC-TA. In particular, we derive the closed-form optimum transmit power and time allocation for the sensor nodes and obtain the optimum UAV's hovering altitude in HA-PC-TA. We reveal the relationship between the optimum parameters and the allowed maximum transmit power of the VAU in UAVPC-TA. Finally, the simulations are conducted to validate the effectiveness of our proposed solutions.

## 1. Introduction

Wireless Body Area Networks (WBANs) have attracted considerable attentions recently for extensive potential applications of developing Internet of Medical Things (IoMT) [1]. In WBANs, the low-power, lightweight and tiny sensor nodes can be implanted under the skin tissue of human body to monitor some physiological parameters such as ECG and EEG [2]. However, the development of WBANs is heavily hindered by the limited energy of sensor nodes [3]. More recently, energy harvesting technology is emerging to supply sufficient energy wirelessly for tiny sensor nodes [4,5]. In particular, sensor nodes can automatically harvest energy from the ambient environment such as sunlight, thermal energy, Radio Frequency (RF) and body movement [6]. It is of great important of ensuring that the sensor nodes can sustainedly obtain energy [7] to work in energy-harvesting WBANs.

RF transmission of Unmanned Aerial Vehicle (UAV) is a promising solution to provide stable RF energy [8] for WBANs outdoors. For example, the organizer would monitor medical data of marathon runners to assess and detect health risk. Here a UAV can be used to energize sensors and collect sensor data. In such UAV-aided WBANs, the UAV acts as an energy source to provide RF energy for sensors and also a mobile relay to receive data from these sensor nodes under the Line of Sight (LoS) communication links [9]. Compared with traditional static access points or base stations, UAV can greatly improve the system performance, such as throughput and reliability by dynamically adjusting its location [10]. However, the WBANs experience interferences from cell networks and satellite communications for these systems may share the same spectrum. For instance, mmWave communications could be adopted in UAV, sensors, cell and satellite [11–13]. Therefore, optimum power control and resource allocation with these interferences taken into consideration should be studied in order to make WBANs sustainable.

Despite great importance, the study mentioned above is overlooked. The authors in Ref. [14] investigated the resource allocation problem with the objective to maximize average throughput for UAV-assisted networks, where the non-convex optimization problem was transformed into a mixed integer nonlinear programming to solve. A novel

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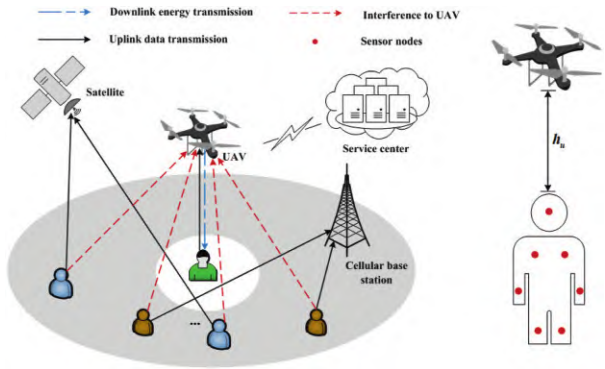


**Fig. 1.** The three-tier hybrid system model.

two-phase resource allocation scheme allocating transmission power, source rate and time slots in order to improve the QoS was proposed in Ref. [15]. For the sum-rate optimization problem in energy-harvesting WBANs, the authors proposed solutions in terms of significance and fairness, power control and power allocation, respectively in Refs. [16–18]. In our previous work [19], an optimal joint power control and time allocation scheme was developed to maximize the sum rate for WBANs with RF energy harvesting. However, none of these works considered the interferences from both cell networks and satellite communications. Although the authors in Ref. [20] considered these interferences, they focused on cooperative beamforming among multiple UAVs to fight against a hostile jammer, which is completely different from our work.

Inspired by the above analysis, this paper is devoted to developing schemes to configure the optimum power control and time allocation and hovering altitude for UAV-aided energy-harvesting WBANs with the objective of maximizing the sum-throughput and the presence of the interferences from other systems. Our main contributions can be summarized as follows.

- First, we consider a harvesting-then-transmitting pattern of sensor nodes in a WBAN coexisting with satellite and macrocell users. We propose two schemes with fixed or channel gain-aware harvesting phase duration, and derive the optimum parameters of sensor nodes and the UAV by formulating non-convex optimization problems which maximize the sum-throughput.
- Second, we decompose the non-convex problem with respect to the first scheme into two tractable subproblems: one is to derive the closed-form optimal transmission power and transmission phase duration of each sensor node; the other is to configure the optimum altitude of the UAV.
- Third, we decompose the non-convex problem with respective to the second scheme into three tractable subproblems, and reveal the relationship of the optimum transmit power of the UAV and time duration of each phase with the allowed maximum transmit power of the UAV.
- Finally, extensive simulations are conducted to evaluate the performance of our proposed schemes. These results confirm the effectiveness of our schemes in terms of improving the sum-throughput with the presence of the mutual interferences.

The remainder of this paper is organized as follows. The system model and problem formulation are detailed in Section 2 and Section 3, respectively. In section 4, we derive the optimal parameters for the first scheme from a perspective of joint UAV hovering altitude, power control and time allocation of sensor nodes. Subsequently, we further show the optimum parameters for the second scheme from a perspective of joint power control of the UAV, and the time allocation of the energy-harvesting phase in Section 5. In Section 6, the simulation results are provided for our proposed solutions, followed by our conclusions in Section 7.

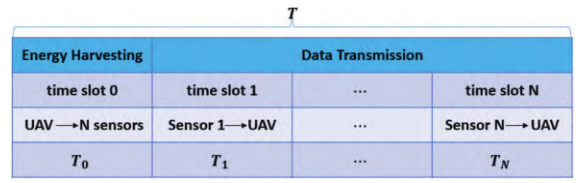


**Fig. 2.** The communication model.

## 2. System model

In this section, we first describe a three-tier hybrid system model, and then the channel and energy model is introduced.

### 2.1. System model

In this paper, we consider a three-tier hybrid system model which consists of a satellite communication network, a cellular communication network and a UAV-aided WBAN, as shown in Fig. 1. These networks overlap and use the same spectrum, which leads to the cross-layer interferences, i.e., the interferences from satellite users and cellular users to the sensors in the UAV-aided WBAN.

The UAV-aided WBAN consists of $N$ sensor nodes, where the UAV is a hovering drone and acts as the RF energy resource and also the sink node to collect data of sensors and transmit them to the service center via an external gateway. We assume that the sensor nodes without battery supply are powered only by the harvested energy from the UAV RF signals [21], and the UAV can be supplied with abundant energy. We consider a two-phase communication model as shown in Fig. 2, where the sensor nodes harvest energy provided by the UAV in the first phase, and then transmit data using the harvested energy to the UAV in a Time Division Multiple Access (TDMA) way in the second phase. After the two phases, the UAV transmits the collected data to the external service center, which is out of the scope of this paper.

More specifically, let the time of one communication window as $T$ which can be normalized to be 1, i.e., $T = 1$, as shown in Fig. 2. The time $T$ is divided into $N+1$ time slots. Note that the duration of each time slot may be unequal. In the time slot $T_0$, sensor nodes can harvest energy from RF radio transmitted by the UAV, i.e. the energy harvesting phase. Then, the $i$-th node utilizes the harvested energy to transmit data to UAV in the $T_i$-th time slot during the data transmission phase where $1 \le i \le N$. In this paper, we consider two cases: (1) the value of $T_0$ is fixed, (2) the value of $T_0$ is configured to maximize the sum-throughput. We would analyze shortly how to set the parameters in the two cases so that the sum-throughput would be maximized.

### 2.2. Channel and energy model

In order to facilitate the analysis, we assume that the hovering altitude of UAV from the human head is $h_u$, as shown in Fig. 1. Denote by $N_s$ and $N_c$ the number of satellite users and cellular users, respectively, and let $j \in \{1, 2, \ldots, N_s\}$ and $k \in \{1, 2, \ldots, N_c\}$. Let $h_{iu}$, $h_{ju}$ and $h_{ku}$ define the channel gains from node $i$, $j$ and $k$ to the UAV, respectively. The channel fading between the UAV and the sensor nodes in a WBAN follows a correlated Rician fading which is composed of LoS and Rayleigh fading components [22,23]. Thus, the channel gain between the UAV and node $i$ will be uniformly expressed as $h_{iu} = (\lambda/4\pi d_{iu})g$ with $g = \bar{g} + \tilde{g}$, where the LoS component $\bar{g}$ depends on the orientation of the UAV and the positions of the UAV and sensors, and the scattering component $\tilde{g}$ follows the i.i.d Rayleigh distribution. $\lambda$ denotes the carrier wavelength, and $d_{iu}$ is the distance between the UAV and node $i$ which can be calculated from



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Euclidean distance. We assume that the carrier wavelength of satellite network, cellular network and WBANs is equal. Similarly, the channel link between the UAV and satellite and cellular users can be described as $h_{ju} = (\lambda/4\pi d_{ju})g$ and $h_{ku} = (\lambda/4\pi d_{ku})g$ respectively, where $d_{ju}$ and $d_{ku}$ express the distance between the UAV and satellite and cellular users.

In the energy-harvesting phase of one communication window $T$, the sensor nodes obtain energy from the RF signal provided by the UAV. Thus, the harvested energy for the node $i$ in $T_0$ is given by

$$E_i = \eta_i P_u h_{ui}^2 T_0 \tag{1}$$

where $\eta_i$ ($0 < \eta_i < 1$) denotes the energy conversion efficiency of the node $i$ and $P_u$ is the transmit power of the UAV. For the sake of simplification, in our model we assume that the link status of the downlink and uplink between the UAV and sensor nodes is invariable in one communication window. That is, the two channels are reversible, i.e., $h_{ui} = h_{iu}$ within $T$.

In the data transmission phase when the UAV collects data from all sensor nodes, the additive white Gaussian noise with the variance of $\delta^2$ is considered. Thus, the Signal-to-Interference-plus-Noise Ratio (SINR) of the UAV from node $i$ can be expressed as

$$\mathrm{SINR}(i) = \frac{P_i h_{iu}^2}{\sum_{j=1}^{N_s} P_j h_{ju}^2 + \sum_{k=1}^{N_c} P_k h_{ku}^2 + \delta^2} \tag{2}$$

where $\sum_{j=1}^{N_s} P_j h_{ju}^2$ and $\sum_{k=1}^{N_c} P_k h_{ku}^2$ are the interferences from the satellite and cellular users sharing the same channel. $P_i$, $P_j$ and $P_k$ denote the transmitting power of node $i$, $j$ and $k$, respectively. More specifically, each node will be allocated a time slot to transmit data to the UAV based on TDMA in the data transmission phase. Thus, the achievable throughput from node $i$ to the UAV can be formulated as the following

$$R_i = \frac{T_i}{T} B \ln(1 + \mathrm{SINR}(i)) \tag{3}$$

where $B$ is the wireless channel bandwidth.

Let $\widehat{E}_i$ denote the energy consumption of node $i$ to transmit data in $T_i$, we have $\widehat{E}_i = \int_0^{T_i} P_i t dt + P_c T_i$, where $P_c$ is the average power consumption of the circuit. As the power of the circuit is much smaller than that of the data transmission, i.e. $P_c \ll P_i$, the energy consumption $\widehat{E}_i$ can be approximated as

$$\widehat{E}_i \approx \int_0^{T_i} P_i t dt \tag{4}$$

## 3. Problem formulation

In this section, we introduce the constraint of the system and present the formulated problem.

### 3.1. Constraint condition

There exist the following constraint conditions.

1) Energy consumption constraint. Since each node can only utilize the harvested energy to transmit data, the energy consumption should not be greater than the harvested energy. For sensor node $i$, we have the following energy constraint $\widehat{E}_i \le E_i$.
2) Power constraint. In the WBAN, the wireless communication generally takes place inside or on the surface of the human body, thus the transmission power of sensor node must be lower than a safety threshold. For node $i$, we have $0 \le P_i \le P_{\max}$.

Similarly, consider the safety factor and the interferences to other users, the UAV who provides the RF signal should have a maximum transmission power limit of $P_{\max}^u$. It can be expressed as follows $0 \le P_u \le P_{\max}^u$.

3) Time allocation constraints. One communication window $T$ is normalized as 1, which includes the energy harvesting phase with time length $T_0$ and the data transmission phase $1 - T_0$. In this paper, we first fix the duration of $T_0$ as a constant of $T/(N+1)$ and then consider a variable $T_0$. In the data transmission phase, each node sends data to the UAV based on TDMA. Thus, the data transmission duration must meet the following requirements $0 \le T_i \le 1$ and $\sum_{i=1}^{N} T_i \le 1 - T_0$.
4) UAV altitude constraint. An appropriate hovering altitude of the UAV is not only ensure the safety but also improve the throughput performance, thus it should be subject to $H_{\min} \le h_u \le H_{\max}$, where $H_{\min}$ and $H_{\max}$ express the minimum and maximum altitude of the UAV, respectively.
5) Interference constraint. Let $I_{\max}$ denote the upper bound of the interferences from $N_s$ satellite users and $N_c$ cellular users. That is, $\sum_{j=1}^{N_s} P_j h_{ju}^2 + \sum_{k=1}^{N_c} P_k h_{ku}^2 \le I_{\max}$.

In what follows, we would first formulate the problem of optimizing the sum-throughput in the first case that $T_0$ is a constant, and would formulate the optimization problem of the second case in Sec. 5.

### 3.2. Problem formulation

Based on Eq. (3), the sum-throughput of all sensor nodes in the WBAN in one communication window can be given by

$$C_o = \sum_{i=1}^{N} R_i = \sum_{i=1}^{N} \frac{T_i}{T} B \ln(1 + \mathrm{SINR}(i)) \tag{5}$$

In the first case, we focus on the design of the hovering altitude of UAV and transmit power and allocated data transmission duration of sensor nodes. To this end, we rewrite the SINR as

$$\mathrm{SINR}(i) \approx P_i \mathscr{F}_i(h_u)$$

where $\mathscr{F}_i(h_u)$ is a simplified formula with the variable $h_u$. Therefore, with the transmit power $P_i$, the time allocation $T_i$ and the UAV altitude $h_u$ as the variables the optimal problem can be formulated as

$$\begin{aligned}
\max_{P_i, T_i, h_u} \quad & C_o = \sum_{i=1}^{N} \frac{T_i}{T} B \ln(1 + P_i \mathscr{F}_i(h_u)) \\
s.t. \quad & C1: \int_0^{T_i} P_i t dt \le E_i \\
& C2: 0 \le P_i \le P_{\max} \\
& C3: 0 \le P_u \le P_{\max}^u \\
& C4: 0 \le T_i \le 1 \\
& C5: \sum_{i=1}^{N} T_i \le 1 - T_0 \\
& C6: H_{\min} \le h_u \le H_{\max} \\
& C7: \sum_{j=1}^{N_s} P_j h_{ju}^2 + \sum_{k=1}^{N_c} P_k h_{ku}^2 \le I_{\max}
\end{aligned} \tag{6}$$

$C1$ is the energy consumption constraint of the sensor nodes, and $C2$ and $C3$ are the power constraints of the sensor nodes and the UAV, respectively. As for the communication model, the time allocation constraints are given by $C4$ and $C5$. Moreover, $C6$ expresses the UAV altitude constraint. Constraint $C7$ limits the interferences from satellite and cellular users. Unfortunately, the Problem (6) with three variables is non-convex. To address this problem, we will develop an efficient solution by transforming it to be two subproblems in the following section.



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## 4. Joint hovering altitude, power control and time allocation with fixed $T_0$

It is prohibitive to find a globally optimal solution to Problem (6), thus a near optimal algorithm is desirable. In this section, we propose a two-step algorithm to solve the optimization problem of the first case from a perspective of joint hovering altitude, power control and time allocation. In the following, we transform the original non-convex problem to two tractable subproblems. First, we fix the UAV hovering altitude $h_u$ and solve the power control and time allocation subproblem using convex optimization theory in Step 1. Then, relying on the result of Step 1, a heuristic iterative algorithm is developed to find the optimal hovering altitude in Step 2.

### 4.1. Transformation of problem (6)

The objective function in Problem (6) is a non-convex and non-concave function with three variables. Thereby, we first fix the UAV hovering altitude and convert the Problem (6) into a power control and time allocation of sensor nodes subproblem. More specifically, for a given hovering altitude, i.e., $h_u = h_0$, the subproblem can be reformulated as

$$\begin{aligned} \Psi 1: \quad & \max_{P_i, T_i} \quad \sum_{i=1}^{N} \frac{T_i}{T} B \ln(1 + P_i \mathscr{T}_i(h_0)) \\ s.t. \quad & C1: \int_0^{T_i} P_i t dt \le E_i \\ & C2: 0 \le P_i \le P_{\max} \\ & C3: 0 \le P_u \le P_{\max}^{u} \\ & C4: 0 \le T_i \le 1 \\ & C5: \sum_{i=1}^{N} T_i \le 1 - T_0 \end{aligned} \tag{7}$$

The subproblem (7) can be transformed to a convex problem, which will be proved in the following Step 1. Thus, for each sensor node, we can obtain the optimum transmission power $P_i^*$ and time allocation duration $T_i^*$. Armed with such results, we would solve the UAV altitude that maximizes the sum-throughput. Correspondingly, the second subproblem can be given by

$$\begin{aligned} \Psi 2: \quad & \max_{h_u} \quad \sum_{i=1}^{N} \frac{T_i^*}{T} B \ln\left(1 + P_i^* \mathscr{T}_i(h_u)\right) \\ s.t. \quad & C6: H_{\min} \le h_u \le H_{\max} \\ & C7: \sum_{j=1}^{N_s} P_j h_{ju}^2 + \sum_{k=1}^{N_c} P_k h_{ku}^2 \le I_{\max} \end{aligned} \tag{8}$$

To solve this problem, we would propose an iterative algorithm in the following Step 2.

### 4.2. Step 1: Transformation of problem $\Psi 1$

Since the Problem $\Psi 1$ is also non-convex, we first transform it into a convex problem. More specifically, let us introduce an auxiliary variable $Z_i = P_i T_i$, and hence the objective function can be converted to $\max_{Z_i, T_i} \sum_{i=1}^{N} \frac{T_i}{T} B \ln(1 + \frac{Z_i \mathscr{T}_i(h_0)}{T_i})$. Correspondingly, the Problem $\Psi 1$ can be rewritten as

$$\begin{aligned} \max_{Z_i, T_i} \quad & \sum_{i=1}^{N} \frac{T_i}{T} B \ln\left(1 + \frac{Z_i \mathscr{T}_i(h_0)}{T_i}\right) \\ s.t. \quad & C1: Z_i \le E_i \\ & C2: 0 \le Z_i \le P_{\max} T_i \\ & C3: 0 \le P_u \le P_{\max}^{u} \\ & C4: 0 \le T_i \le 1 \\ & C5: \sum_{i=1}^{N} T_i \le 1 - T_0 \end{aligned} \tag{9}$$

In the following, we introduce Lemma 1 and Lemma 2 to show the concavity of the objective function in Problem (9).

**Lemma 1.** *Assume that $f(x) = a \ln(1 + bx)$ is a function of $x \in [0, X]$, where $a$ and $b$ are constants greater than 0. We introduce a variable $z$ as $z = tx$ with $t \in [0, 1]$. Then, $m(t, z) = tf(z/t)$ is concave in $(t, z)$ when $t \in [0, 1]$ and $\forall z \in [0, tX]$.*

Proof. We can derive the second derivative of $f(x)$ as $f''(x) = \frac{-ab^2}{(1+bx)^2}$. For $f''(x) \le 0$, $f(x)$ is a concave function. Then, we calculate the Hessian matrix of $m(t, z)$ as

$$\nabla^2 m(t, z)) = \frac{f''(z/t)}{t^3} \begin{bmatrix} z^2 & -zt \\ -zt & t^2 \end{bmatrix}$$

For $\forall x, y \in \mathbf{R}$ and $t \in [0, 1]$, we further have

$$[x \quad y] \nabla^2 m(t, z) \begin{bmatrix} x \\ y \end{bmatrix} = \frac{f''(z/t)}{t^3} (zx - yt)^2 \le 0$$

Hence, the Hessian matrix $\nabla^2 m(t, z)$ is a negative semi-definite matrix. Therefore, $m(t, z) = tf(z/t)$ is a concave function [24].

**Lemma 2.** *Let $\widehat{g}(x_i, t_i) = a x_i \ln\left(1 + \frac{b t_i}{x_i}\right)$, $(i = 1, 2, \ldots, N)$, where $a$ and $b$ are constants which is not smaller than 0, then $G(x, t) = \sum_{i=1}^{N} \widehat{g}(x_i, t_i)$ is a concave function.*

*Proof.* Let $f$: $R^N \to R$, we denote the mapping function as

$$f(k) = e^T k$$

where

$$e = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

The mapping function $f(k)$ can reduce the $N$-dimensional set to the one-dimemsional set.

Moreover, let $g$: $R^2 \to R^N$, the corresponding mapping function that can improve the two-dimensional set to $N$-dimensional set is defined as

$$g(x, t) = \begin{bmatrix} (x, t) \\ g_2(x, t) \\ \vdots \\ g_N(x, t) \end{bmatrix} = \begin{bmatrix} \widehat{g}(x_1, t_1) \\ \widehat{g}(x_2, t_2) \\ \vdots \\ \widehat{g}(x_N, t_N) \end{bmatrix}$$

Furthermore, we define the composite function $G(x, t)$ as



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$$G(x,t) = f(g_1(x,t), g_2(x,t), \ldots, g_N(x,t))$$
$$= [1,1,\ldots,1]\begin{bmatrix}(x,t)\\ g_2(x,t)\\ \vdots\\ g_N(x,t)\end{bmatrix} = [1,1,\ldots,1]\begin{bmatrix}\widehat{g}(x_1,t_1)\\ \widehat{g}(x_2,t_2)\\ \vdots\\ \widehat{g}(x_N,t_N)\end{bmatrix}$$

According to Lemma 1, we can know that $\widehat{g}(x_i,t_i)$ is a concave function. For any $i = 1, 2, \ldots, N$, the second derivative of $\widehat{g}(x_i,t_i)$ is not greater than 0, i.e. $\widehat{g}^{''}(x_i,t_i) \le 0$. Correspondingly, the second derivative of $G(x, t)$ can be calculated as

$$G^{''}(x,t) = g^{''}(x,t)^T \nabla^2 f(g(x,t)) g^{'}(x,t) + \nabla f(g(x,t))^T g^{''}(x,t)$$

Since $f(k)$ is a linear function and the Hessian matrix of $f$ is nondecreasing in each argument, we get that $G''(x, t)$ is less than or equal to 0. As a consequence, we can prove that $G(x, t)$ is a concave function [24]. □

Following Lemma 1 and Lemma 2, we have that the objective function of Problem (9) is a concave function. According to the convex optimization theory [24], a problem to find the maximum of a concave function is a convex optimization problem. Hence, we address the Problem (9) by convex optimization in the following subsection.

### 4.3. Step 1: The solution to problem Ψ1

Here we address the power control and time allocation of sensor nodes by decoupling Problem (9) into two more tractable subproblems. The first one is to optimize the data transmission time duration $T_i$ in data transmission phase. For a given altitude $h_0$, we have

$$\begin{aligned} T_i^*(Z_i) = \quad & \underset{T_i}{\arg\max} \sum_{i=1}^{N} \frac{T_i}{T} B\ln\left(1 + \frac{Z_i \mathscr{F}_i(h_0)}{T_i}\right) \\ s.t. \quad & 0 \le T_i \le 1 \\ & \sum_{i=1}^{N} T_i \le 1 - T_0 \end{aligned} \tag{10}$$

Correspondingly, the second subproblem is to optimize the auxiliary variable $Z_i$, which can be formulated as

$$\begin{aligned} Z_i^* = \quad & \underset{Z_i}{\arg\max} \sum_{i=1}^{N} \frac{T_i^*}{T} B\ln\left(1 + \frac{Z_i \mathscr{F}_i(h_0)}{T_i^*}\right) \\ s.t. \quad & Z_i \le E_i \\ & 0 \le Z_i \le P_{\max}(1 - T_0) \end{aligned} \tag{11}$$

#### 4.3.1. The solution to problem (10)

In order to convert the inequality constraints of problem (10) into the equality constraints, we first introduce the following Lemma 3.

**Lemma 3.** *If $P_i^*$ and $T_i^*$ are the optimal solutions to problem (7), there exists $T_i^*$ establishing $\sum_{i=1}^{N} T_i^* = 1 - T_0$.*

*Proof.* Assume that $P_i^*$ and $T_i^*$ are the optimal solutions to problem (7), and $T_i^*$ satisfies $\sum_{i=1}^{N} T_i^* < 1 - T_0$. Thus, there must exist a constant $c$ that establishes $\sum_{i=1}^{N} T_i^* + c = 1 - T_0$. Let $P_i^*$ and $T_i^* + c$ be the suboptimal solutions to the problem (7), it holds that

$$\sum_{i=1}^{N} \frac{T_i^*}{T} B\ln\left(1 + P_i^* \mathscr{F}_i(h_0)\right) < \sum_{i=1}^{N} \frac{T_i^* + c}{T} B\ln\left(1 + P_i^* \mathscr{F}_i(h_0)\right)$$

This contradicts the assumption that $P_i^*$ and $T_i^*$ are the optimal solutions. Therefore, if $P_i^*$ and $T_i^*$ are the optimal solutions to problem (7), $T_i^*$ must satisfy $\sum_{i=1}^{N} T_i^* = 1 - T_0$.

According to Lemma 3, the inequality constraint of problem (10) can be reasonably transformed into the equality constraint $\sum_{i=1}^{N} T_i = 1 - T_0$. Thus, we can use the Lagrange multiplier method [24] to solve the convex optimization problem (10). More specifically, by introducing a nonnegative multiplier $\mu$, we can obtain that

$$W = \sum_{i=1}^{N} \frac{T_i}{T} B\ln\left(1 + \frac{Z_i}{T_i} \mathscr{F}_i(h_0)\right) - \mu\left(\sum_{i=i}^{N} T_i - 1 + T_0\right) \tag{12}$$

Then the first derivative of Eq. (12) can be calculated as

$$W^{'} = \sum_{i=1}^{N} \frac{1}{T} B\ln\left(1 + \frac{Z_i}{T_i} \mathscr{F}_i(h_0)\right) - \sum_{i=1}^{N} \frac{BZ_i \mathscr{F}_i(h_0)}{T_i + Z_i \mathscr{F}_i(h_0)} - \sum_{i=1}^{N} \mu \tag{13}$$

By setting Eq. (13) to zero, i.e., let $W' = 0$, for $\forall i \in \{1, 2, \ldots, N\}$, we can get

$$\mu = B\ln\left(1 + \frac{Z_i}{T_i} \mathscr{F}_i(h_0)\right) - \frac{BZ_i \mathscr{F}(h_0)}{T_i + Z_i \mathscr{F}_i(h_0)}$$

Let $r(x) = B\ln(1 + x) - \frac{Bx}{(1+x)} (x \ge 0, B \ge 0)$, we can derive the first derivative of $r(x)$ as

$$r^{'}(x) = \frac{B}{1+x} - \frac{B}{(1+x)^2} = \frac{Bx}{(1+x)^2} \ge 0$$

demonstrating that $r(x)$ is a monotonically increasing function with respect to $x$.

Let $x = (Z_i \mathscr{F}_i(h_0))/T_i$, we can get

$$L = \frac{Z_1 \mathscr{F}_1(h_0)}{T_1} = \frac{Z_2 \mathscr{F}_2(h_0)}{T_2} = \cdots = \frac{Z_N \mathscr{F}_N(h_0)}{T_N} \tag{14}$$

Transforming Eq. (14) yields

$$\begin{cases} T_1 L = Z_1 \mathscr{F}_1(h_0) \\ T_2 L = Z_2 \mathscr{F}_2(h_0) \\ \quad \cdots \\ T_N L = Z_N \mathscr{F}_N(h_0) \end{cases}$$

We can further obtain that $L = \frac{\sum_{i=1}^{N} (\mathscr{F}_i(h_0) Z_i)}{1 - T_0}$.

As a result, combining with $L = (Z_i \mathscr{F}_i(h_0))/T_i$, we can eventually get

$$T_i^* = \frac{(1 - T_0) \mathscr{F}_i(h_0) Z_i}{\sum_{i=1}^{N} (\mathscr{F}_i(h_0) Z_i)} \tag{15}$$

#### 4.3.2. The solution to problem (11)

Substituting the optimal solution $T_i^*$ into Eq. (15), the problem (11) can be rewritten as

$$\begin{aligned} Z_i^* = \quad & \underset{Z_i}{\arg\max} (1 - T_0) B\ln\left(1 + \frac{\sum_{i=1}^{N} (\mathscr{F}_i(h_0) Z_i)}{1 - T_0}\right) \\ s.t. \quad & Z_i \le E_i \\ & 0 \le Z_i \le P_{\max}(1 - T_0) \end{aligned} \tag{16}$$

Let $\mathrm{m}(Z_i) = (1 - T_0) B\ln\left(1 + \frac{\sum_{i=1}^{N} (\mathscr{F}_i(h_0) Z_i)}{1 - T_0}\right)$, the first derivative of $m(Z_i)$ can be derived as

$$\mathrm{m}^{'}(Z_i) = \frac{\sum_{i=1}^{N} \mathscr{F}_i(h_0)(1 - T_0) B}{(1 - T_0) + \sum_{i=1}^{N} \mathscr{F}_i(h_0) Z_i} \ge 0$$

Consequently, $m(Z_i)$ is monotonically increasing, and the optimal solution of problem (11) satisfies

$$\begin{aligned} Z_i^* &= \min(P_{\max}(1 - T_0), E_i) \\ &= \min\left(P_{\max}(1 - T_0), \eta_i P_u h_{ui}^2 T_0\right) \end{aligned} \tag{17}$$



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Therefore, we obtain the optimum transmission power

$$P_i^* = \frac{Z_i^*}{T_i^*} = \min\left(\frac{P_{\max}(1-T_0)}{T_i^*}, \frac{\eta_i P_u h_{ui}^2 T_0}{T_i^*}\right) \tag{18}$$

In summary, we can have the optimal transmission power and time allocation duration for each sensor node in data transmission phase by merging Eq. (15) and Eq. (18).

### 4.4. Step 2: The solution to problem Ψ2

As mentioned in Step 1, we fix the UAV hovering altitude to optimize the transmission power and allocation time for each sensor node, denoted as $P_i^*$ and $T_i^*$ for $i = 1, 2, \ldots, N$. In this subsection, we try to determine the optimal UAV hovering altitude in order to maximize the sum-throughput relying on the results obtained in Step 1. Substituting the optimal $P_i^*$ and $T_i^*$ into the Problem Ψ2, we can derive the sum-throughput of all sensor nodes as $\sum_{i=1}^{N} \frac{T_i^*}{T} B\ln(1 + P_i^* \mathscr{T}_i(h_u))$. According to the property of the Log function, we can find that the greater the value of $\mathscr{T}_i(h_u)$, the greater the sum-throughput. Therefore, Problem Ψ2 can be approximatively simplified as

$$\begin{aligned} \max_{h_u} \quad & \sum_{i=1}^{N} \mathscr{T}_i(h_u) \\ s.t. \quad & C6: H_{\min} \le h_u \le H_{\max} \\ & C7: \sum_{j=1}^{N_s} P_j h_{ju}^2 + \sum_{k=1}^{N_c} P_k h_{ku}^2 \le I_{\max} \end{aligned} \tag{19}$$

**Algorithm 1:** A heuristic iterative algorithm to find the optimum hovering altitude

**Input:** $H_{\min}, H_{\max}, I_{\max}, \varepsilon$
**Output:** $h_u^*$
Initialize $h_u^0 = H_{min}$
Calculate $L_{max} = \lfloor (H_{\max} - H_{\min})/\varepsilon \rfloor$
**for** $(n = 0; n < L_{max} - 1; n++)$ **do**
  Calculate $I(h_u^n) = \sum_{j=1}^{N_s} P_j h_{ju}^2 + \sum_{k=1}^{N_c} P_k h_{ku}^2$
  **if** $I(h_u^n) \le I_{\max}$ **then**
    Calculate $F(h_u^n) = \sum_{i=1}^{N} \mathscr{T}_i(h_u^n)$
  **end**
  $h_u^{n+1} = h_u^n + \varepsilon$
**end**
$h_u^* = arg\max(F(h_u^n))$
return $h_u^*$

Combining Eq. (2) and Eq. (3) with Eq. (3.2), we know that the objective function $\sum_{i=1}^{N} \mathscr{T}_i(h_u)$ is a really complicated quadratic nonlinear function. Moreover, the constraint $C7$ in problem (19) is also quadratic nonlinear with the variable $h_u$. The optimal solution is thus intractable to be achieved. Hence, we propose a heuristic iterative algorithm by constituting an arithmetic progression of $h_u$ ranging from $H_{\min}$ to $H_{\max}$ to address the problem (19). Let $\epsilon$ denote the update step length of the UAV hovering altitude. A formal description of the heuristic iterative algorithm to optimize the hovering altitude is shown in Algorithm 1.

### 4.5. The solution to the original problem

In Section 4.2, 4.3 and 4.4, we have studied the power control and time allocation subproblem in Step 1 and the optimal hovering altitude subproblem in Step 2, respectively. In this subsection, we combine these two steps to jointly optimize the UAV hovering altitude and transmission power and time allocation duration of sensor nodes in order to maximize the sum-throughput of all sensor nodes in the WBAN.

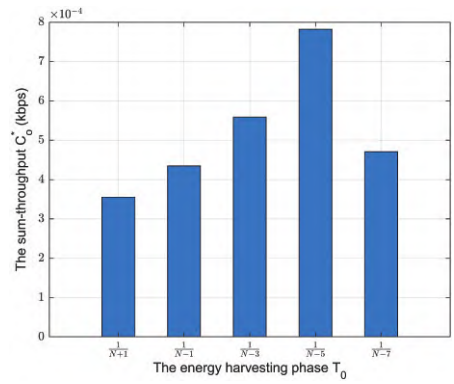


**Fig. 3.** The sum-throughput with energy harvesting duration $T_0$.

More specifically, for the $n$-th iteration, the optimum transmission power and time allocation $\{P_i^*, T_i^*\}^{(n)}$ can be obtained in Step 1 with a fixed $h_u^{(n-1)}$. Then, relying on these results, we can obtain the optimal UAV hovering altitude $h_u^{*(n)}$ in Step 2. Thus, the pseudo-optimal sum-throughput $C_o^{(n)}(P_i^*, T_i^*, h_u^*)$ can be achieved based on Eq. (5). After that, we update the UAV hovering altitude in the ($n$+1)-th iteration as $h_u^{*(n)}$, and conduct the optimization in Step 1 and Step 2 based on the latest $h_u^{*(n)}$. These procedures repeat until the difference of the sum-throughput between two successive iterations is less than a threshold. Let Θ be the iterative stopping threshold. If the following condition is satisfied

$$C_o^{(n+1)}(P_i^*, T_i^*, h_u^*) - C_o^{(n)}(P_i^*, T_i^*, h_u^*) \le \Theta$$

the optimal sum-throughput in the WBAN can be given by

$$C_o^* \triangleq C_o^{(n+1)}(P_i^*, T_i^*, h_u^*) \tag{20}$$

where the optimum transmission power and time allocation of sensor nodes are given by $\{P_i^*, T_i^*\} \triangleq \{P_i^*, T_i^*\}^{(n+1)}$, and the optimal hovering altitude is $h_u^* \triangleq h_u^{*(n+1)}$. The procedure of the two-step optimization algorithm for the joint Hovering Altitude, Power Control and Time Allocation (HA-PC-TA) is summarized in Algorithm 2.

**Algorithm 2:** A two-step algorithm: HA-PC-TA

Initialize a feasible $h_u^{(1)}$, a threshold Θ
Initialize $n = 0$, $C_o^{(0)} = 0$
**repeat**
  update iteration indicator $n = n + 1$
  Get $\{P_i^*, T_i^*\}^{(n)}$ based on Eq.(15) and Eq.(18)
  Get $h_u^{*(n)}$ according to Algorithm 1
  Calculate $C_o^{(n)}(P_i^*, T_i^*, h_u^*)$ based on Eq.(5)
  Update $h_u^{(n+1)} = h_u^{*(n)}$
**until** It satisfies $C_o^{(n)} - C_o^{(n-1)} \le \Theta$
Set $C_o^* \triangleq C_o^{(n)}(P_i^*, T_i^*, h_u^*)$
Set $\{P_i^*, T_i^*\} \triangleq \{P_i^*, T_i^*\}^{(n)}$
Set $h_u^* \triangleq h_u^{*(n)}$

### 4.6. Algorithm implementation

In this subsection, we evaluate the performance of the sum-throughput with the different energy harvesting phase duration $T_0$, which is illustrated in Fig. 3. In order to optimize the data transmission duration $T_i$ for each sensor node, we fix the duration of the energy harvesting phase. Here we let $T_0 = \frac{1}{N+1}$. Let $N = 10$ and $P_u = 20$dBm, we find in Fig. 3 that the sum-throughput can achieve an optimal value when $T_0$ is not equal to $\frac{1}{11}$. More specifically, the maximum value of the sum-



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throughput reaches to $7.82 \times 10^{-4}$ kbps as $T_0 = \frac{1}{N-5}$, which outperforms the suboptimal value as $T_0 = \frac{1}{N+1}$ over up to 119.7%. However, the sum-throughput is dramatically reduced by 39.9% when $T_0$ continuously increases to $\frac{1}{N-7}$. This observation demonstrates that it is necessary to further optimize the energy harvesting phase duration $T_0$ to achieve the maximum sum-throughput. Hence, we proceed to the optimization problem in the second case with a variable $T_0$.

## 5. Joint UAV power control and time allocation with variable $T_0$

Inspired by the observations above, we further study the UAV power control and time allocation problem in this section.

### 5.1. Problem formulation

In the UAV-aided energy-harvesting WBAN, the transmission power $P_i$ in data transmission phase is determined by the harvested energy $E_i$ in the energy-harvesting phase for each sensor node. Moreover, the harvested energy $E_i$ depends on the transmit power $P_u$ of the UAV and the duration $T_0$ of the energy-harvesting phase according to Eq. (1). Assume that the harvested energy during the energy-harvesting phase for each sensor node must be all used for data transmission in the second phase, it thus holds that $P_i = E_i/T_i$. Based on Eq. (5), given the output hovering altitude $h_u^*$ of the UAV from Algorithm 2, the joint UAV power control and time allocation problem can be formulated as

$$\begin{aligned} \max_{P_u, T_0, T_i} \quad & C_o' = \sum_{i=1}^{N} \frac{T_i}{T} B\ln\left(1 + \frac{P_u T_0 \mathscr{R}(i)}{T_i}\right) \\ s.t. \quad & C1: 0 \le P_u \le P_{\max}^u \\ & C2: 0 \le P_u \le \frac{P_{\max}(1-T_0)}{T_0 \sum_{i=1}^{N} \eta_i h_{ui}^2} \\ & C3: 0 \le T_0 \le 1 \\ & C4: 0 \le T_i \le 1 \\ & C5: \sum_{i=1}^{N} T_i \le 1 - T_0 \end{aligned} \tag{21}$$

where $\mathscr{R}(i) = \frac{\eta_i h_{ui}^2 h_{lu}^2}{\sum_{j=1}^{N_s} P_j h_{ju}^2 + \sum_{k=1}^{N_c} P_k h_{ku}^2 + \delta^2}$.

More specifically, in Problem (21), $C1$ and $C2$ are the UAV's power constraints, and $C3$, $C4$ and $C5$ are the time allocation constraints. Note that Problem (21) is also non-convex. However, it is reasonably proved that the objective function is a concave function according to Lemma 1 and Lemma 2. Accordingly, we will develop an efficient solution by transforming the Problem (21) to be three convex subproblems to solve [25].

The first one is to optimize data transmission time $T_i$ for each sensor node in the data transmission phase. For given $P_u$ and $T_0$, it can be expressed as

$$\begin{aligned} T_i^*(P_u, T_0) = \quad & \arg\max_{T_i} \sum_{i=1}^{N} \frac{T_i}{T} B\ln\left(1 + \frac{P_u T_0 \mathscr{R}(i)}{T_i}\right) \\ s.t. \quad & C4: 0 \le T_i \le 1 \\ & C5: \sum_{i=1}^{N} T_i \le 1 - T_0 \end{aligned} \tag{22}$$

Correspondingly, given $T_0$ and the optimized $T_i^*(P_u, T_0)$, the second subproblem is to optimize the UAV's power $P_u$ in the energy-harvesting phase. Thus, we have

$$\begin{aligned} P_u^*(T_0) \quad & = \arg\max_{P_u} \sum_{i=1}^{N} \frac{T_i^*(P_u, T_0)}{T} B\ln\left(1 + \frac{P_u T_0 \mathscr{R}(i)}{T_i^*(P_u, T_0)}\right) \\ s.t. \quad & C1: 0 \le P_u \le P_{\max}^u, \\ & C2: 0 \le P_u \le \frac{P_{\max}(1-T_0)}{T_0 \sum_{i=1}^{N} \eta_i h_{ui}^2} \end{aligned} \tag{23}$$

Subsequently, we can derive the third optimization subproblem to derive the optimum energy-harvesting phase duration $T_0$ with the optimum solution $T_i^*(P_u^*(T_0), T_0)$ and $P_u^*(T_0)$ given. We have $T_0^* =$

$$\begin{aligned} \arg\max_{T_0} \quad & \sum_{i=1}^{N} \frac{T_i^*(P_u^*(T_0), T_0)}{T} B\ln\left(1 + \frac{P_u^*(T_0) T_0 \mathscr{R}(i)}{T_i^*(P_u^*(T_0), T_0)}\right) \\ s.t. \quad & C2: 0 \le P_u \le \frac{P_{\max}(1-T_0)}{T_0 \sum_{i=1}^{N} \eta_i h_{ui}^2} \\ & C3: 0 \le T_0 \le 1 \end{aligned} \tag{24}$$

Obviously, by solving the three convex subproblems (22), (23) and (24) respectively, we can obtain the optimal solution $T_i^*(P_u, T_0)$, $P_u^*(T_0)$ and $T_0^*$. In what follows, we solve the above three subproblems in sequence.

### 5.2. The solution to subproblem (22)

We solve Subproblem (22) using the same approach as [26]. First, we introduce a non-negative Lagrange multiplier $\lambda$ associated with the constraint in Subproblem (22). Thus, the Lagrangian can be presented as

$$G = \sum_{i=1}^{N} \frac{T_i}{T} B\ln\left(1 + \frac{P_u T_0 \mathscr{R}(i)}{T_i}\right) - \lambda\left(\sum_{i=1}^{N} T_i - 1 - T_0\right)$$

Then, for each $i \in \{1, 2, \ldots, N\}$, by setting the first derivative of $G$ with respect to $T_i$ equal to zero, we can have

$$\lambda = \frac{1}{T}\left(B\ln\left(1 + \frac{P_u T_0 \mathscr{R}(i)}{T_i}\right) - \frac{B P_u T_0 \mathscr{R}(i)}{T_i + P_u T_0 \mathscr{R}(i)}\right) \tag{25}$$

Let $m = P_u T_0 \mathscr{R}(i)/T_i$, Eq. (25) will be converted to

$$\lambda = \frac{1}{T} B\ln(1+m) - \frac{Bm}{1+m} \tag{26}$$

It is easily verified that Eq. (26) is an increasing function with respect to $m$. Therefore, there must exist a unique $m$ which makes Eq. (26) true. Since it holds that

$$m = \begin{cases} P_u T_0 \mathscr{R}(1)/T_1, i = 1 \\ P_u T_0 \mathscr{R}(2)/T_2, i = 2 \\ \cdots \\ P_u T_0 \mathscr{R}(N)/T_N, i = N \end{cases}$$

and $\sum_{i=1}^{N} T_i = 1 - T_0$, we can get

$$T_i^*(P_u, T_0) = \frac{(1-T_0) P_u T_0 \mathscr{R}(i)}{\sum_{i=1}^{N} (P_u T_0 \mathscr{R}(i))} \tag{27}$$

### 5.3. The solution to subproblem (23)

Inserting $T_i^*(P_u, T_0)$ into the objective function of Subproblem (23) and combining the two constraints $C1$ and $C2$, we can rewrite the Subproblem (23) as

$$\begin{aligned} P_u^*(T_0) \quad & = \arg\max_{P_u} (1-T_0) B\ln\left(1 + \frac{\sum_{i=1}^{N} P_u T_0 \mathscr{R}(i)}{1-T_0}\right) \\ s.t. \quad & 0 \le P_u \le \min(P_{\max}^u, P_{th}(1-T_0)/T_0) \end{aligned} \tag{28}$$



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where $P_{th} = P_{\max}/\sum_{i=1}^{N}\eta_i h_{ui}^2$. It can be easily verified that the optimal function in Subproblem (28) is an increasing function of $P_u$. Thus, the optimal solution is obtained at its maximum value as $P_u^*(T_0) = \min(P_{\max}^u, P_{th}(1-T_0)/T_0)$.

### 5.4. The solution to subproblem (24)

It is observed that $P_u^*(T_0)$ is chosen from two possible values. Therefore, we can solve Subproblem (24) by substituting $P_u^*(T_0)$ and $T_i^*(P_u, T_0)$ into it and separating it into two problems, each of which chooses one possible value of $P_u^*(T_0)$. Specifically, let $P_u^*(T_0) = P_{\max}^u$, we can have

$$\begin{aligned} T_0^{'} &= \arg\max_{T_0} (1-T_0)B\ln\left(1+\frac{\sum_{i=1}^{N}P_{\max}^u T_0\mathscr{R}(i)}{1-T_0}\right) \\ s.t. \quad & 0 \le T_0 \le \frac{P_{th}}{P_{th}+P_{\max}^u} \end{aligned} \tag{29}$$

Correspondingly, let $P_u^*(T_0) = \frac{P_{th}(1-T_0)}{T_0}$, we then get

$$\begin{aligned} T_0^{''} &= \arg\max_{T_0} (1-T_0)B\ln\left(1+\sum_{i=1}^{N}P_{th}\mathscr{R}(i)\right) \\ s.t. \quad & \frac{P_{th}}{P_{th}+P_{\max}^u} \le T_0 \le 1 \end{aligned} \tag{30}$$

In the following, we will solve problem (29) and problem (30) in sequence.

#### 5.4.1. The solution to problem (29)

First, let $f(T_0) = (1-T_0)B\ln(1+QT_0/(1-T_0))$, where $Q = \sum_{i=1}^{N}P_{\max}^u\mathscr{R}(i)$. Then, the second derivative of $f(T_0)$ can be derived as

$$-\frac{Q^2}{(1+QT_0-T_0)^2(1-T_0)} \le 0$$

indicating that $f(T_0)$ is a concave function. Thereby, the optimal solution to Problem (29) is unique, and we can solve it by setting the first derivative to be zero:

$$-\ln\left(1+\frac{QT_0}{1-T_0}\right)+\frac{Q}{1+(Q-1)T_0} = 0 \tag{31}$$

The solution to Eq. (31) can be obtained by the bisection method. Denote $T_0^{'*}$ as the value of $T_0$ got from Eq. (31). Then, we can have $T_0^{'} = \max(0, \min(T_0^{'*}, P_{th}/(P_{th}+P_{\max}^u)))$.

#### 5.4.2. The solution to problem (30)

It's clear that the objective function in Problem (30) is a monotonically decreasing function with respect to $T_0$. Therefore, it reaches the maximum value when $T_0 = \frac{P_{th}}{P_{th}+P_{\max}^u}$.

According to the convex optimization theory [24], after solving the above two subproblems (29) and (30) respectively, the solutions to the subproblems are the optimal solutions to Problem (24). Hereby, the three convex optimal subproblems (22), (23) and (24) have been solved, respectively. Correspondingly, the optimal solutions to subproblems (22), (23) and (24).

In conclusion, the optimization problem of maximizing the sum-throughput of all sensor nodes in one communication window can be solved by jointly tuning the UAV transmission power $P_u$ and the time allocation $T_0$ and $T_i$. A formal description of our solution is stated in Algorithm 3 named UAVPC-TA.

**Table 1**
Simulation parameters.

| | |
|---|---|
| The energy conversion efficiency $\eta_i$ | 0.9 |
| The maximum transmission power $P_{\max}$ | 12 dBm |
| The number of sensor nodes $N$ | 10 |
| The minimum UAV hovering height $H_{\min}$ | 5 m |
| The maximum UAV hovering height $H_{\max}$ | 25 m |
| Background noise $\delta^2$ | −40 dBm |

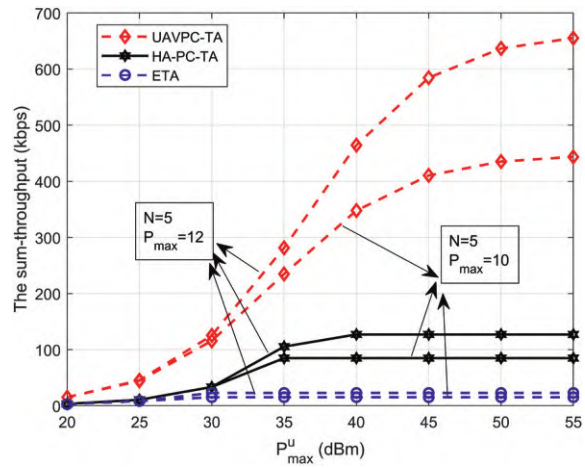


**Fig. 4.** The sum-throughput vs. $P_{max}^u$ when $N = 5$.

**Algorithm 3:** UAVPC-TA: Joint UAV Power Control and Time Allocation Algorithm

Let $P_u^*, T_0^*, T_1^*, \ldots, T_N^*$ denote the optimal solutions
Initialize $T_0^{'} = T_0^{''} = R1 = R2 = 0$
Solve subproblem (29) and get
$T_0^{'} = \max(0, \min(T_0^{'*}, \frac{P_{th}}{P_{th}+P_{\max}^u}))$
Let $R1 = (1-T_0^{'})B\ln(1+\frac{\sum_{i=1}^{N}P_{\max}^u T_0\mathscr{R}(i)}{1-T_0^{'}})$
Solve subproblem (30) and get
$T_0^{''} = \frac{P_{th}}{P_{th}+P_{\max}^u}$
Let $R2 = (1-T_0^{''})B\ln(1+\sum_{i=1}^{N}P_{th}\mathscr{R}(i))$
**if** $R1 > R2$ **then**
  $T_0^* = T_0^{'}$ and $P_u^* = P_{\max}^u$
**else**
  $T_0^* = T_0^{''}$ and $P_u^* = \frac{P_{th}(1-T_0)}{T_0}$
**end if**
**for** $i = 1$ to $N$ **do**
  $T_i^* = \frac{(1-T_0^*)P_u^* T_0^*\mathscr{R}(i)}{\sum_{i=1}^{N}(P_u^* T_0^*\mathscr{R}(i))}$
**end for**

## 6. Simulation results

In this section, we will evaluate the performance of our proposed schemes and compare them with the existing scheme ETA that averages each time slot, i.e. $T_0 = T_1 = \cdots = T_N$. It should be noted that our first scheme that is to jointly optimize the UAV hovering height, sensor nodes power control and time allocation is named by HA-PC-TA, and our second scheme that is to jointly optimize the UAV power and time allocation is referred to as UAVPC-TA. Moreover, three kinds of users are located in an area of 30 m × 30 m. $N_s = 10$ satellite users and $N_c = 10$ cellular users are randomly distributed in the area. For simplicity, the transmission power of satellite users and cellular users is taken the same value



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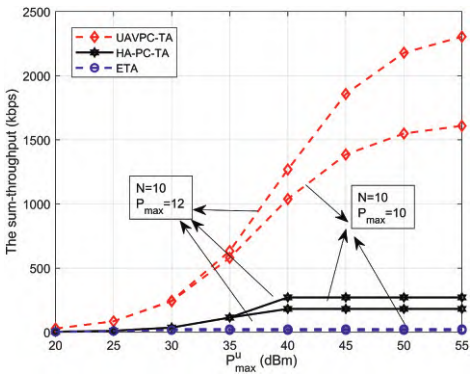


**Fig. 5.** The sum-throughput vs. $P^u_{max}$ when $N = 10$.

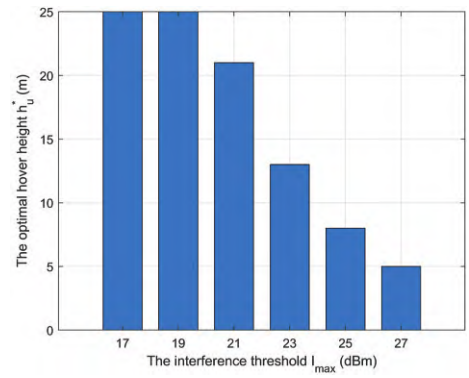


**Fig. 7.** The optimal hover height $h^*_u$ vs. $I_{\max}$.

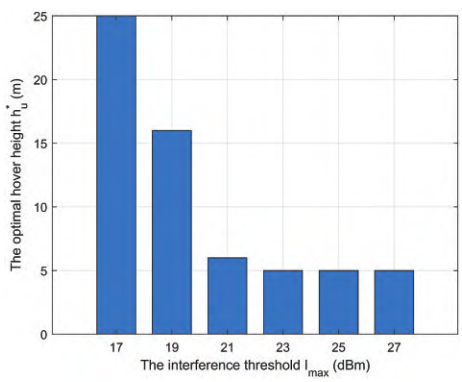


**Fig. 6.** The optimal hover height $h^*_u$ vs. $I_{\max}$.

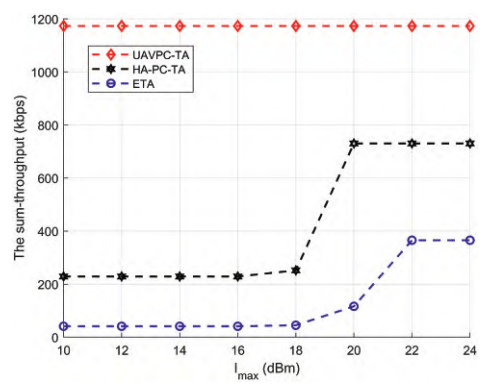


**Fig. 8.** The sum-throughput of all sensor nodes vs. $I_{\max}$.

respectively, i.e. $P_j = 30$ dBm and $P_k = 20$ dBm. Furthermore, the carrier frequency is 2.4 GHz and the channels between the UAV and the users follow the combination of LoS and Rayleigh fading, where the fading parameter is $g = 10$ dB. Other main simulation parameters are shown in Table 1.

We first evaluate the impact of the maximum transmission power of the UAV on the sum-throughput of all sensor nodes when $N = 5$, as shown in Fig. 4. We can observe from it that the sum-throughput of the scheme UAVPC-TA, HA-PC-TA and ETA increases with the increase of the maximum transmission power of the UAV. Moreover, among the three schemes, the scheme UAVPC-TA shows the best sum-throughput. Specifically, the sum-throughput of the scheme UAVPC-TA is 415.6% higher than the scheme HA-PC-TA and 2775.4% higher than scheme ETA, When $P^u_{max} = 55$ dBm and $P_{\max} = 12$ dBm. This is because that the scheme UAVPC-TA adapts the energy harvesting time $T_0$ to the UAV transmission power $P_u$. And all the energy harvested by the sensor nodes is used for data transmission. But in scheme HA-PC-TA and ETA, the energy harvested by sensor nodes is the same, but the former optimizes the data transmission time and transmission power of sensor nodes, while the latter only optimizes the transmission power of sensor nodes. When $P^u_{max}$ is very small, there is little difference in the optimum transmission power and data transmission time among the three schemes, due to the low energy harvested by the sensor nodes. However, with the increase of $P^u_{max}$, the energy harvested by the sensor nodes increases, and the difference of the optimum transmission power and data transmission time among the three schemes increases gradually.

Correspondingly, Fig. 5 depicts the sum-throughput of all sensor nodes when $N = 10$. It further demonstrates the superiority of our UAVPC-TA scheme in terms of the sum-throughput. Through the longitudinal comparison between Figs. 4 and 5, it can be seen that $P_{\max}$ has a relatively large influence on the sum-throughput of all sensor nodes. A larger $P_{\max}$ leads to a bigger sum-throughput. This can be interpreted as follows. When $P_{\max}$ is very small, it is highly likely that $p^*_u = P_{th}(1 - T_0)/T_0$. As $P_{\max}$ is very large, it is highly likely that $p^*_u = P^u_{\max}$. In addition, $P^u_{\max}$ and $P_{th}(1 - T_0)/T_0$ are independently valued, so the sum-throughput of all sensor nodes is also independent.

Fig. 6 describes the variation of the optimum hovering altitude of the UAV with interference threshold $I_{\max}$ and $N_s = 6$ and $N_c = 6$. Obviously, the optimum hovering height of the UAV decreases with the increase of $I_{\max}$. When $I_{\max}$ exceeds 23 dBm, the optimum hovering height of the UAV does not change. This is mainly because when $I_{\max}$ constraints are relatively strict, the UAV needs to narrow the hovering height value range to meet the constraints. When the constrain $I_{\max}$ is relatively loose, the UAV can achieve optimum hover height within $H_{\min}$ and $H_{\max}$. Similarly, Fig. 7 illustrates the variation of optimum hover height with interference threshold $I_{\max}$ and $N_s = 10$ and $N_c = 10$. Through the comparative analysis of Figs. 6 and 7, we observe that $N_s$ and $N_c$ have a great impact on the UAV hovering altitude. This is mainly because the bigger values of $N_s$ and $N_c$ means more interferences, making UAV have to hover at a higher altitude.

Fig. 8 shows the sum-throughput of all sensor nodes with diverse $I_{\max}$. Without loss of generality, we set $N = 5$, $N_c = 6$ and $N_s = 6$. Obviously our proposed UAVPC-TA scheme performs best in terms of the sum-throughput of all sensor nodes. And we can find that the sum-throughput of the HA-PC-TA and ETA schemes increases gradually With the increase of $I_{\max}$. The main reasons is that as the interference constraint $I_{\max}$ is strict, the optimum hovering altitude of the UAV has to be increased so that the sum-throughput will not achieve the optimum value. Once the interference constraint is relatively loose, the UAV can reduce the optimum hover altitude to achieve the optimal sum-throughput. In the UAVPC-TA scheme, the optimum hovering altitude of the UAV has a great influence on energy acquisition and data transmission of sensor nodes, which makes the performance of UAVPC-TA scheme more stable. In the HA-PC-TA and ETA schemes, the optimum



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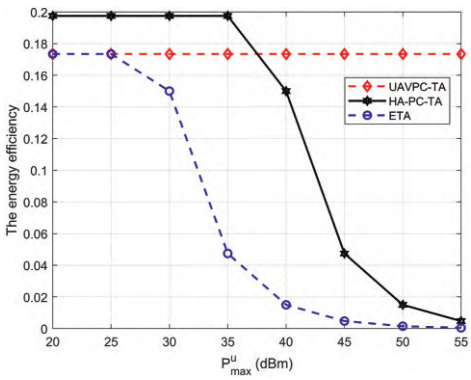


**Fig. 9.** The energy efficiency vs. $P^u_{max}$

hovering height of the UAV mainly affects the data transmission of sensor nodes.

Fig. 9 shows the energy efficiency with diverse maximum transmission power when the sum-throughput achieves the optimum one. The energy efficiency is defined as $e_k = \frac{\sum_{i=1}^{N} P_i T_i}{P_u T_0}$. We observe that with the increase of $P^u_{max}$, the energy efficiency of the UAVPC-TA scheme remains stable, that of the HA-PC-TA scheme first remains constant and then decreases, and that of the ETA scheme decreases gradually. The main reason is that all the harvested energy of sensor nodes in the UAVPC-TA scheme is used for data transmission during the energy harvesting time $T_0$, however, that is fixed in the HA-PC-TA scheme, and that is overlooked in ETA scheme.

## 7. Conclusions

In this paper, we have studied joint power control and time allocation schemes to maximize the sum-throughput of UAV-aided energy harvesting WBANs. Based on a three-tier hybrid network model and a two-phase communication model, an optimization problem with the objective to maximize the sum-throughput of all sensor nodes was formulated. Then, a joint UAV hovering altitude, power control and time allocation of sensor nodes solution was proposed, which was transformed to two subproblems to solve by convex optimization theory and a heuristic iterative algorithm. Subsequently, we further developed a joint UAV power control and time allocation of the energy harvesting and data transmission scheme which was converted to three convex optimization subproblems to solve. Finally, simulation results demonstrated that our proposed schemes performed better in terms of the sum-throughput.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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